XUEWU CAI

FINANCIAL ENGINEERING & MORTGAGE ANALYTICS LEADER

Clarksville, MD | 410-446-6815 | gourmetcai@gmail.com

linkedin.com/in/xuewu-cai-a8069664 | U.S. citizen

PROFESSIONAL SUMMARY

Financial engineering director and technology leader with more than 20 years across mortgage finance, quantitative research, risk management and trading systems. Experience includes market-data infrastructure and analytical reporting, enterprise software and financial model implementation, documented use of generative AI tools in engineering workflows. Combines hands-on Java, Python, MATLAB and C/C++ expertise with large-scale analytics on AWS and IBM Symphony and documented financial model implementation.

CORE EXPERTISE

Finance & risk: Credit risk transfer (CRT), structured finance, mortgage analytics, portfolio risk, capital planning, ROE, CECL, DFAST, derivatives pricing, scenario analysis, and stress testing. Mortgage products: MBS, TBA, agency debt, and CMO; hedge accounting.

Leadership & delivery: People management, technical leadership, multi-project delivery, financial model implementation, and investment decision support.

Technology: Java, Python, MATLAB, C/C++, AWS, IBM Symphony, Docker, IntexCalc, DealMaker, QuantLib, FIX, Oracle, SQL Server, Snowflake, MongoDB, Spring Boot, PyTorch, JSON, VS Code, Amazon Q, Copilot, and harness engineering.

PROFESSIONAL EXPERIENCE

Freddie Mac | Director, Financial Engineering/Inv&Capital Market

May 2014 – Present | McLean, VA

- Own an enterprise forecasting platform for income statements, balance sheets, and CECL estimates, supporting annual business planning, risk appetite assessment, capital planning, and DFAST stress testing.
- Lead financial engineering projects supporting STACR bond issuance and ACIS reinsurance within Freddie Mac’s CRT (Credit Risk Transfer) program, which protects more than $1.6 trillion in mortgage loans against credit losses.
- Direct development of modeling tools that combine prepayment and default models, large-scale computing on AWS and IBM Symphony, and Intex valuation analytics to assess bond performance and transaction economics, including return on committed capital.
- Delivered deal memorandum, capital, economics, and ROE analytics for new RPL and synthetic structures, using Copilot and harness engineering to support delivery.
- Use generative AI tools, including Copilot, extensively to streamline engineering workflows and improve delivery efficiency.
- Manage the transition from LIBOR to SOFR within CRT modeling tools and analytics.
- Oversee portfolio execution and risk analytics covering duration, convexity, option-adjusted spread (OAS), key rate durations (KRDs), capital benefits, post-transaction capital, and return on equity across the CRT portfolio.
- Apply hands-on expertise in STACR, senior/subordinate structures, ACIS reinsurance, and IntexCalc, DealMaker, and CDI structures to support valuation and model implementation.

PROFESSIONAL EXPERIENCE — CONTINUED

Campbell & Company | Research Engineer and Engineering Roles

March 2003 – April 2014 | Washington, DC–Baltimore Area

Research Engineer, Portfolio and Risk Team | 2009 – 2014

- Developed volatility and correlation forecasting models in MATLAB using daily, intraday, implied volatility, and tick data.
- Applied statistical methods, Monte Carlo simulation, and bootstrapping to investigate indicators of market stress, portfolio drawdowns, regime changes, and correlation breakdowns using MATLAB, C/C++, and Java.
- Designed and implemented internal risk reporting covering factor analysis, scenarios, stress tests, duration, and margin to support Investment Committee decisions.
- Produced regulatory and industry reporting for SEC, CFTC, and OPERA requirements, as well as marketing and compliance analyses, using MATLAB, Java/StyleReport, and Panopticon.

Quantitative Finance Engineer | 2006 – 2008

- Led development of a Java-based intraday cash equity trading system using the Caché database; built time-series data infrastructure and market data feeds in MFC/C++.
- Designed a Java connectivity layer that unified MATLAB access to production databases, including Oracle, SQL Server, Caché, and QADirect.
- Built an FX options pricing, simulation, and risk engine using Visual C++, Boost, and QuantLib.

Senior Software Engineer | 2003 – 2006

- Led architecture and development of a multithreaded, fault-tolerant electronic trading platform using Java, Portware, and FIX, supporting concurrent strategies, orders, and multiple asset classes.
- Supported more than $3 billion in capital on the trading platform and helped save tens of millions of dollars in slippage costs.

Aquilent | Senior Software Engineer

June 1999 – February 2003

- Built Java/Linux sensor integration systems, including embedded Linux deployments, JNI hardware interfaces, serial communications, XML instrument definitions, and dynamic sensor discovery.
- Designed server-side satellite data management and visualization components, XML data models, and a JavaScript-based model equation engine.
- Developed Apache/Tomcat visualization portals, session administration tools, and an RMI socket solution enabling callbacks through enterprise firewalls.

EDUCATION & PROFESSIONAL DEVELOPMENT

University of Maryland Baltimore County — Master’s Degree, Computer Science

Shanghai Jiao Tong University — M.S., Computer Science & Engineering

Dalian University of Technology — Bachelor’s Degree, Applied Mathematics

Professional development: CAIA (Chartered Alternative Investment Analyst) Level I

LANGUAGES

Chinese: native or bilingual proficiency | English: full professional proficiency